

Supporting Evidence

Buried Rescue · Track 1 (Ideas) · Ochiai, Ojiro · every figure below can be reproduced from public data with the linked code

1. THE HEADLINE: EVERY ATTACHMENT OPENED

NCI-funded open-access papers, 2016–2022, with attachments and no data-repository accession	Count
Papers checked (Europe PMC; no text-mined accession for GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB, GenBank)	26,228
Papers with a spreadsheet attachment · all their attachments downloaded and opened	4,759 · 25,411
Real datasets found* (in 3,736 papers, about 1 in 7 of all papers checked; 85% openly licensed)	9,543
...with no permanent identifier of their own (excluding ~2,200 in BMC and PLOS journals, which publishers already mirror to Figshare)	7,335
NCI grants credited by the 3,736 papers holding datasets · their FY2016–2022 awards (NIH RePORTER)	3,598 · \$10.3B
Still happening: 2024–2025 papers doing the same (600-paper sample)	23.2%
Not just cancer: NHLBI 14.9% · NIDDK 17.6% · NIMH 18.7% · NIA 20.5% · NINDS 23.8% (NCI 17.9%); 124,758 papers across all 24 institutes	NIH-wide

*Dataset = largest table has at least 50 rows × 3 columns, or 1,000 rows. The strict rule leaves out small tables like the numbers behind one chart.

2. CHECKING THE METHOD: FOUR RANDOM SAMPLES OF 300 PAPERS

Papers holding a real dataset: glioblastoma 19.6% · breast 18.8% · colorectal 19.4% · leukemia 17.2%. All four together: 202 of 1,076 papers (18.8%); 156 (14.5%) as spreadsheets.

The four cancers agree within three points; the spreadsheet rate (14.5%) matches the full corpus (14.9%). **Blind human checks:** 60 random files judged by hand without seeing the program's answers (two sets of 30): 54 of 60 agree (90%; Cohen's kappa 0.83; set 1 26/30, set 2 28/30). In 3 of 5 disagreements the program was stricter (tables of 43–44 rows). An earlier check of 90 files with the AI assistant: 60/60 "dataset" calls real. On 500 random papers, 15% mention a repository the text-mining missed.

3. TWO EXAMPLES YOU CAN OPEN

- **121-patient brain tumor cohort** — *Acta Neuropathol Commun* 2018, PMC6193287. Publisher Figshare link (10.6084/m9.figshare.7222268) credits a different funder; absent from NCI's index.
- **118,273-row drug screen** — *Nat Commun* 2019, PMC6599391. No identifier on Figshare or DataCite.

4. RESCUED DATA PUT BACK TO WORK

- Two rescued brain tumor tables from different papers join on 244 shared patients (over 96% agreement). Pooled to 1,037 patients, they reproduce the known IDH survival difference (validation, not a new finding).
- Sample patient tables: ~4,400 patients, ~500 from hospital cohorts not in TCGA, GEO or dbGaP.
- **Existing systems checked:** BioStudies, publisher Figshare deposits, NLM FAIR-SMART, NLM Dataset Catalog. None sorts datasets by content, counts by funder, or reaches NCI's index.

5. BUILT AND READY

The cancer data is in the attachments.

For years, NCI-funded scientists shared data by attaching spreadsheets to papers. Those files exist, but no catalogue lists them. **This tool searches all of them, opens any one in a click, and shows what any NCI grant left behind.**

Search the buried data

Show me a random verified dataset

15,458

buried spreadsheet files

4,759

papers, 2016–2022

9,371

real dataset spreadsheets, every file opened (9,543 counting tables inside PDF and Word files)

82%

under CC BY or CC0

18.8 GB

of Excel and CSV

Search buried files... try: survival, proteomics, breast, methylation, cohort

PMC6193287

or an NCI grant, e.g. P30CA0166

Any year

Any license

Verified first, then largest

Verified only

Clear

2 files · 1 paper

- **Catalogue file:** all 9,543 datasets in the Index of NCI Studies' exact 27-column format (deliverable/). Eight also packaged with DataCite metadata for deposit.
- **Working tool:** attachment-rescue.vercel.app/app — search, look up any paper or NCI grant, open the file.
- **Code and data (MIT):** github.com/OJ-IRO/buried-rescue

Cited: FAIR-SMART (PMID 41066526) · Gobeill 2025 (PMC12371329) · Federer 2018, 2022 · Colavizza 2020 · Ghosh 2025 (PMC12782234) · Boja 2026 (PMC13479720) · Boyd 2026 (JNCI djag199) · CBIIT INS-Data.